so, strangely enough, string theory claims that there is no difference whatsoever between a “fat” Garden-hose universe and a “thin” one.

It’s a cosmic hedging of bets, somewhat akin to what you, as a smart investor, should do if faced with the following puzzle. Imagine you learn that the fate of two stocks trading on Wall Street—say, a company making fitness machines and a company making heart-bypass valves—are inextricably connected. They each closed trading today valued at one dollar per share, and you are told by a reliable source that if one company’s stock goes up the other’s will go down, and vice versa. Moreover, your source—who is completely trustworthy (but whose guidance might be crossing over legal boundaries)—tells you that the next day’s closing prices of these two companies are absolutely certain to be inversely related to one another. That is, if one stock closes at $2 per share, the other will close at $½ (50 cents) per share; if one stock closes at $10 per share, the other will close at $1/10 (10 cents) per share, and so on. But the one thing your source can’t tell you is which stock will close high and which will close low. What do you do?

Well, you immediately invest all of your money in the stock market, equally divided between the shares of these two companies. As you can easily check by working out a few examples, no matter what happens on the next day, your investment cannot lose value. At worse it can remain the same (if both companies again close at $1), but any movement of share prices—consistent with your insider information—will increase your holdings. For instance, if the fitness company closes at $4 and the heart-valve company closes at $¼ (25 cents), their combined value is $4.25 (for each pair of shares), compared with $2 the previous day Furthermore, from the perspective of net worth, it does not matter one bit whether the fitness company closes high and the heart-valve company low, or vice versa. If you care only about the total amount of money, these two distinct circumstances are financially indistinguishable.

The situation in string theory is analogous in that the energy in string configurations comes from two sources—vibrations and windings—whose contributions to the total energy of a string are generally different. But, as we shall see in more detail below, certain pairs of distinct geometrical circumstances—leading to high-winding-